Logical Coherence Score: Experimental Evaluation

FactReasoner — LCS experiments

July 14, 2026

1 Setup

This report evaluates the Logical Coherence Score (LCS) pipeline over 9 worked examples from `data/lcs`, across 3 model(s) (llama-4-maverick-17b, llama-3.3-70b-instruct, gpt-oss-120b), 3 conditional-strength uncertainty-quantification method(s) (surr-lp, surr-smp, verbal). Relations are mined with the **win** candidate-pair policy (order window $w = 4$). Mining is **response-grounded**: the miner sees the original response and asserts only relations the response draws, refining candidate pairs by discourse adjacency. For every mined coherence MRF all four LCS readouts are computed: mean_marginal, consistency, reified, log_partition.

2 Dataset

Table 1: Per-example size: atoms and mined relations. The **Code** column is the short label used in the per-score tables.

Example	Code	Atoms	Relations
aeroparts	`aero`	16	18 (1.1)
ex1-damages	`e1-dmg`	13	18 (1.4)
ex2-biography	`e2-bio`	19	30 (1.6)
ex2-biography-contradicted	`e2-con`	12	5 (0.4)
ex3-narrative	`e3-nar`	33	36 (1.1)
ex4-summary	`e4-sum`	15	34 (2.3)
ex5-renda-K	`e5-K`	15	38 (2.5)
ex5-renda-S	`e5-S`	18	31 (1.7)
ex6-incident	`ex6-incident`	13	26 (2.0)

3 Results

4 Findings

Averaged over all model/example cells, the headline mean-marginal LCS by conditional-strength method is: surr-lp=0.738, surr-smp=0.708, verbal=0.732. Differences here reflect how each UQ method sets the edge strengths that drive the coherence MRF; the verbalized baseline is included for comparison.

Contradiction-heavy examples average mean-marginal 0.582 versus 0.767 for the cleaner ones, consistent with the LCS being pulled down when the response asserts a live internal conflict.

Table 2: LCS score **mean_marginal**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.73	0.91	0.91	0.49	0.53	0.50	0.55	0.41	0.95
	surr-smp	0.78	0.79	0.91	0.44	0.51	0.51	0.53	0.36	0.94
	verbal	0.73	0.83	0.87	0.51	0.72	0.63	0.87	0.79	0.93
llama-3.3-70b-instruct	surr-lp	0.70	0.89	0.97	0.81	0.72	0.33	0.65	0.56	0.95
	surr-smp	0.72	0.87	0.97	0.19	0.70	0.29	0.71	0.54	0.95
	verbal	0.68	0.68	0.96	0.50	0.67	0.55	0.72	0.59	0.94
gpt-oss-120b	surr-lp	0.74	0.85	0.88	0.58	0.79	0.77	0.94	0.87	0.94
	surr-smp	0.67	0.88	0.88	0.65	0.79	0.77	0.91	0.90	0.94
	verbal	0.63	0.85	0.65	0.45	0.70	0.71	0.85	0.83	0.93

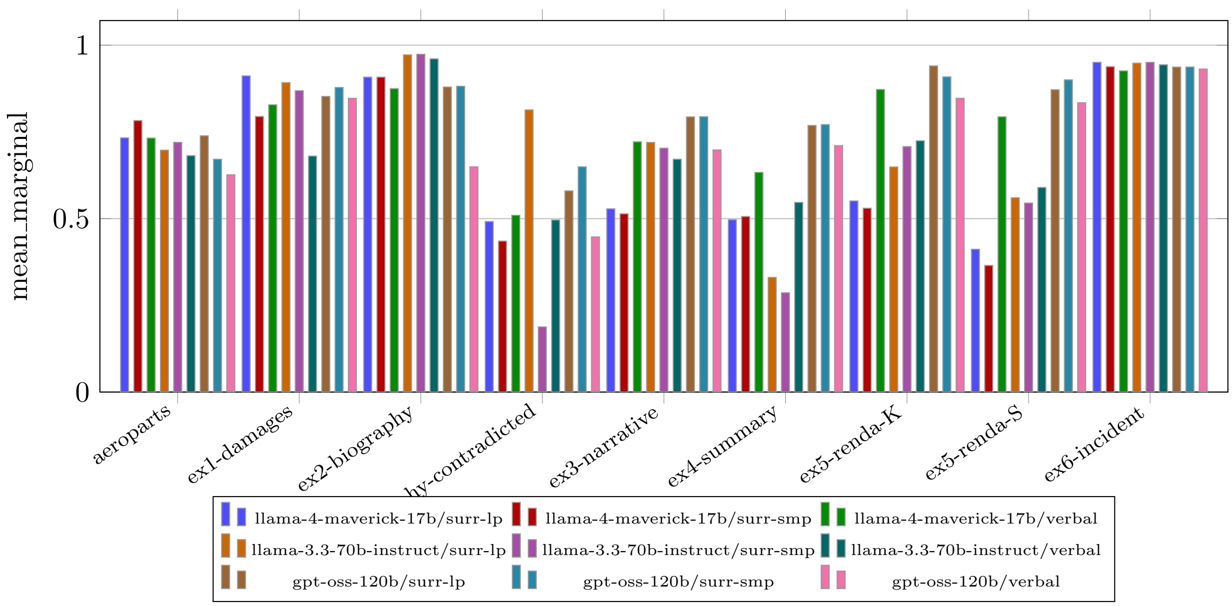


Figure 1: mean_marginal by example, per model and conditional-strength UQ method.

Table 3: LCS score **consistency**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	Config	aero	e1-dmg	e2-bio	e2-con	e3-nar	e4-sum	e5-K	e5-S	ex6-incident
llama-4-maverick-17b	surr-lp	0.85	0.00	1.00	0.90	1.00	0.04	1.00	0.06	1.00
	surr-smp	0.67	0.00	1.00	0.78	1.00	0.01	1.00	0.05	1.00
	verbal	0.83	0.03	1.00	0.92	0.60	0.63	0.02	0.00	1.00
llama-3.3-70b-instruct	surr-lp	0.55	0.07	1.00	0.00	0.07	0.19	1.00	0.01	1.00
	surr-smp	0.52	0.01	1.00	0.05	0.00	0.43	1.00	0.01	1.00
	verbal	0.60	0.03	1.00	0.43	1.00	0.11	0.27	0.51	1.00
gpt-oss-120b	surr-lp	0.16	0.00	1.00	0.06	0.00	0.43	0.00	0.00	1.00
	surr-smp	0.21	0.00	1.00	0.05	0.09	1.00	0.00	0.00	1.00
	verbal	0.44	0.00	1.00	0.55	0.10	1.00	0.00	0.00	1.00

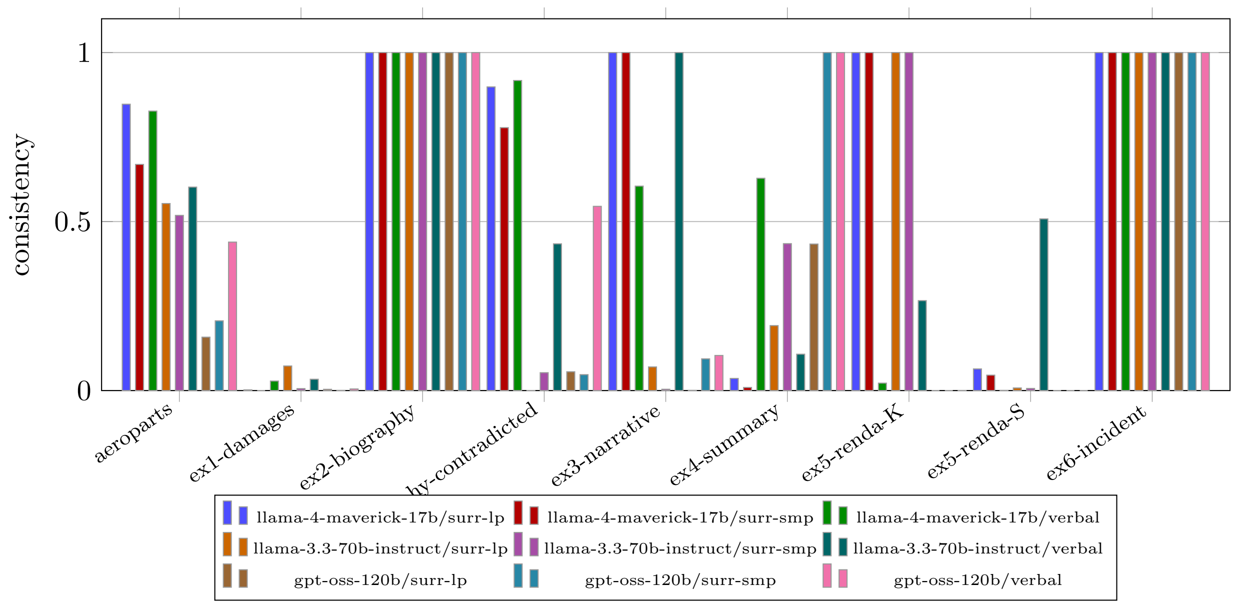


Figure 2: consistency by example, per model and conditional-strength UQ method.

Table 4: LCS score **reified**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	**Config**	**aero**	**e1-dmg**	**e2-bio**	**e2-con**	**e3-nar**	**e4-sum**	**e5-K**	**e5-S**	**ex6-incident**
llama-4-maverick-17b	surr-lp	0.47	0.50	0.50	0.50	0.30	0.50	0.49	0.43	0.50
	surr-smp	0.46	0.50	0.50	0.50	0.50	0.50	0.42	0.50	0.50
	verbal	0.42	0.20	0.47	0.46	0.18	0.37	0.01	0.05	0.50
llama-3.3-70b-instruct	surr-lp	0.43	0.38	0.50	0.02	0.48	0.24	0.50	0.31	0.50
	surr-smp	0.44	0.17	0.50	0.47	0.43	0.50	0.50	0.31	0.50
	verbal	0.37	0.29	0.46	0.48	0.38	0.19	0.13	0.08	0.49
gpt-oss-120b	surr-lp	0.50	0.50	0.50	0.50	0.50	0.44	0.50	0.50	0.50
	surr-smp	0.46	0.47	0.50	0.36	0.50	0.50	0.50	0.47	0.50
	verbal	0.40	0.15	0.18	0.37	0.32	0.44	0.27	0.21	0.50

Table 5: LCS score **log_partition**: rows are (model, pair policy, conditional-strength UQ method); columns are the examples (short codes; full names in Table 1). Higher is more coherent.

Model	**Config**	**aero**	**e1-dmg**	**e2-bio**	**e2-con**	**e3-nar**	**e4-sum**	**e5-K**	**e5-S**	**ex6-incident**
llama-4-maverick-17b	surr-lp	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00	1.00
	surr-smp	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00	1.00
	verbal	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00
llama-3.3-70b-instruct	surr-lp	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00
	surr-smp	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00
	verbal	0.00	0.00	1.00	0.00	1.00	0.00	0.00	0.00	1.00
gpt-oss-120b	surr-lp	0.00	0.00	1.00	0.00	0.00	0.00	0.00	0.00	1.00
	surr-smp	0.00	0.00	1.00	0.00	0.00	1.00	0.00	0.00	1.00
	verbal	0.00	0.00	1.00	0.00	0.00	1.00	0.00	0.00	1.00

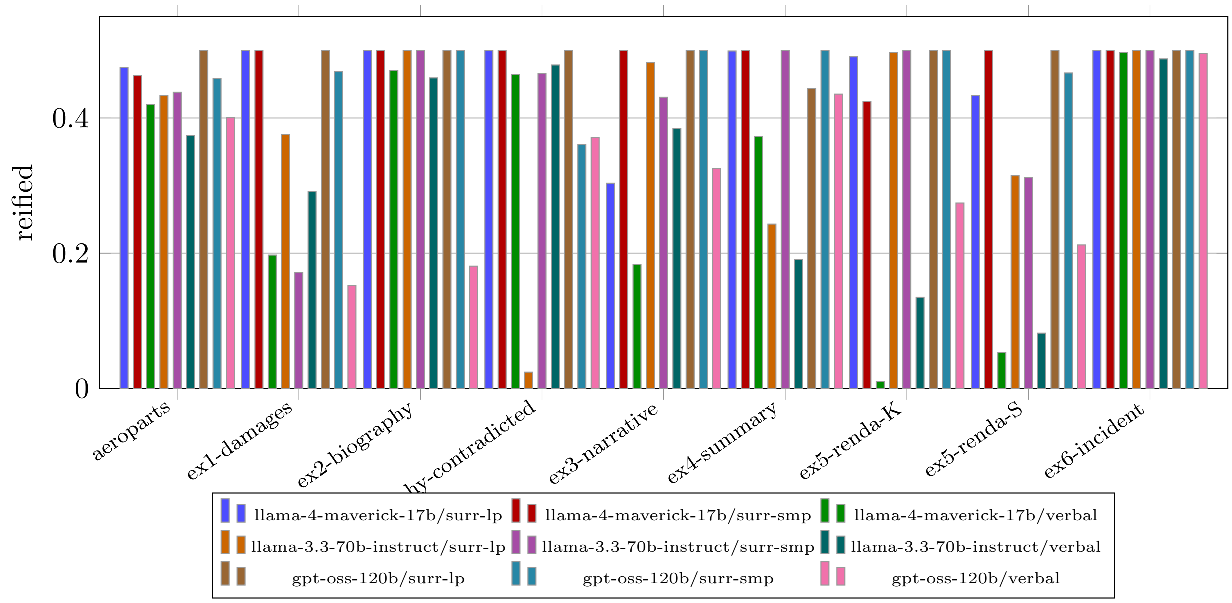


Figure 3: reified by example, per model and conditional-strength UQ method.

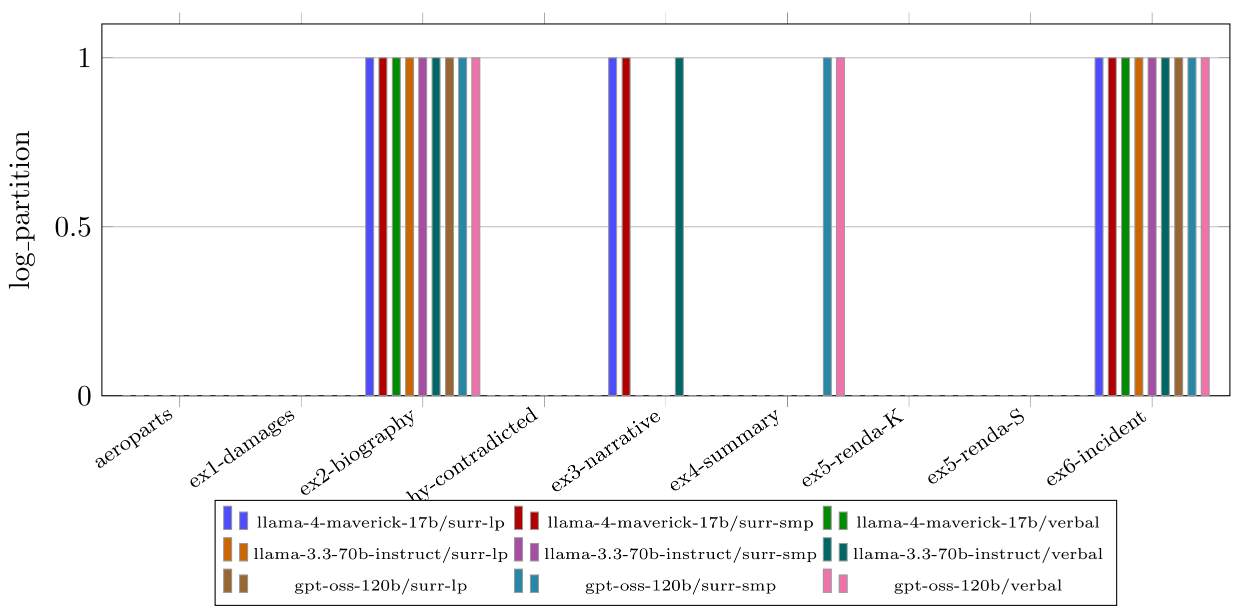


Figure 4: log_partition by example, per model and conditional-strength UQ method.

Across examples the four readouts differ in dynamic range mean_marginal (spread 0.79), consistency (spread 1.00), reified (spread 0.49), log_partition (spread 1.00); **consistency** is the most discriminative in this run.

5 Case study: an authored coherent response

Design. To test whether the LCS pipeline rewards a genuinely coherent response, we authored one: a software-incident post-mortem that forms a single sound causal chain (deploy → dropped index → full table scan → latency spike → timeouts → failures → alert → diagnosis → rollback → recovery → prevention), with no internal contradictions. It was run through the same matrix as the other examples: three models, three conditional-strength UQ methods, and the windowed candidate-pair policy with response-grounded mining.

Result. The example scores as coherent: mean-marginal averages 0.940, and crucially the consistency readout averages 1.000 — i.e. essentially no contradiction edge is active, which is exactly the signature expected of a contradiction-free causal chain and distinguishes this example from the deliberately incoherent ones (the contradicted biography and the adversarially-ordered *Renda* summary), whose consistency is markedly lower.

Graph density. Under this pipeline the mined graph stays sparse (1.9 relations per atom on this 13-atom response), close to what the response actually asserts rather than the dense web of competing constraints that all-pairs mining produces (and that collapses the marginal-based scores toward the prior even for a coherent response). The mined graphs for this example appear in Section 7.

Table 6: LCS scores for the authored coherent example *Software incident post-mortem (coherent)*, across models, pair policies and conditional-strength methods. `r/at` is relations-per-atom (graph density).

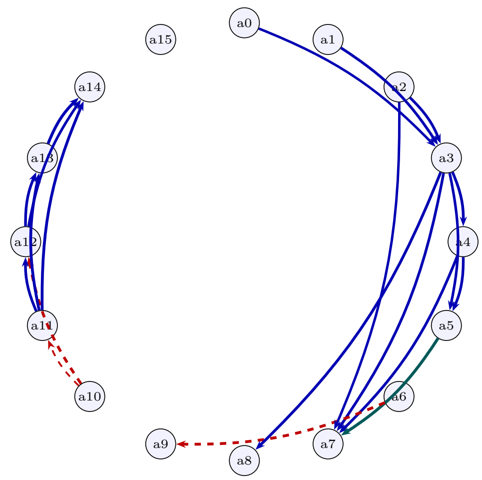
Model	**Pol.**	**Grd.**	**Strength**	`r/at`	mean	consist	reified	logZ
llama-4-maverick-17b	win	g	surr-lp	2.0	0.95	1.00	0.50	-8.3
	win	g	surr-smp	1.8	0.94	1.00	0.50	-8.7
	win	g	verbal	1.8	0.93	1.00	0.50	-9.6
llama-3.3-70b-instruct	win	g	surr-lp	1.7	0.95	1.00	0.50	-8.2
	win	g	surr-smp	1.8	0.95	1.00	0.50	-8.2
	win	g	verbal	1.8	0.94	1.00	0.49	-9.8
gpt-oss-120b	win	g	surr-lp	2.0	0.94	1.00	0.50	-8.0
	win	g	surr-smp	1.9	0.94	1.00	0.50	-8.0
	win	g	verbal	2.0	0.93	1.00	0.50	-10.2

6 Threats to validity

Over-connection is controlled in this run. Because relations were mined with the windowed policy and response-grounded prompts, the graphs stay sparse (mean 1.7 relations per atom, 20% labelled contradiction) — in the plausible range for coherent prose rather than the dense web of competing constraints that all-pairs mining produces. The marginal-based readouts can therefore be read directly here. The residual threat is the opposite one: windowing plus grounding can *miss* a genuine relation the response draws, so recall of long-range links (only partially recovered by the discourse-adjacency gate) remains the main quantity to validate against human-labeled relations.

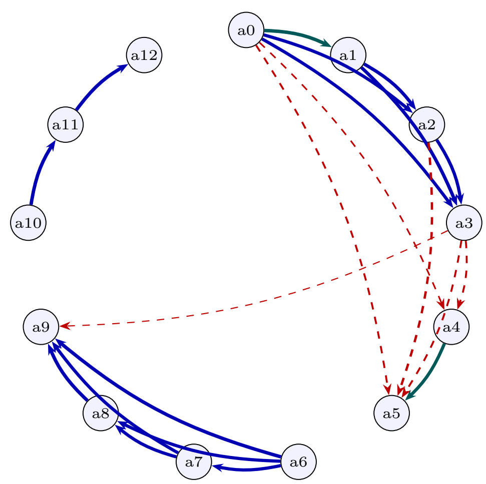
7 Relation graphs

Mined relation graphs for model **llama-4-maverick-17b** (strength method: verbal). Nodes are atoms on a circle; edges are mined relations — solid blue = entailment, dashed red = contradiction, solid teal = equivalence — with thickness proportional to the mined probability. Graphs use the win candidate-pair policy with response-grounded mining, which keeps the graph close to what the response actually asserts.



win: 20 relations

Figure 5: Relation graph for `aero` (AeroParts turbine-blade recall report): win.



win: 21 relations

Figure 6: Relation graph for `e1-dmg` (Legal damages paragraph): win.

8 Conclusion

The experiment harness runs the full LCS pipeline end to end — atom-pair relation mining, coherence Markov-random-field construction, and all four score readouts (mean marginal support, consistency probability, reified coherence node, and normalized log-partition) — across multiple

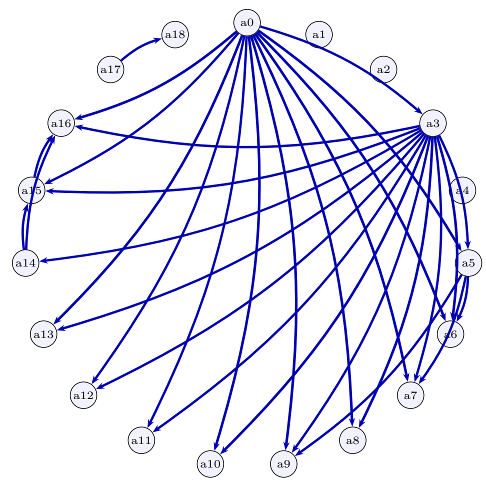


Figure 7: Relation graph for `e2-bio` (Biography (consistent)): win.

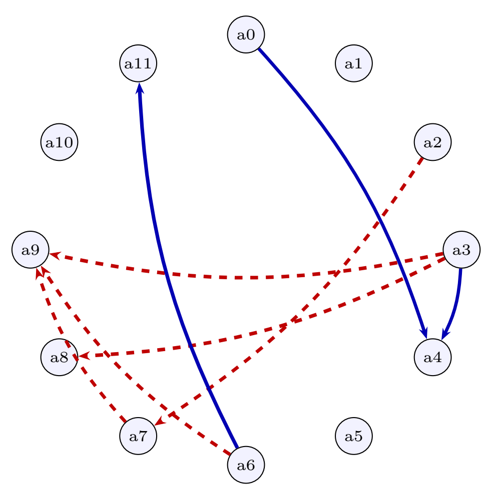
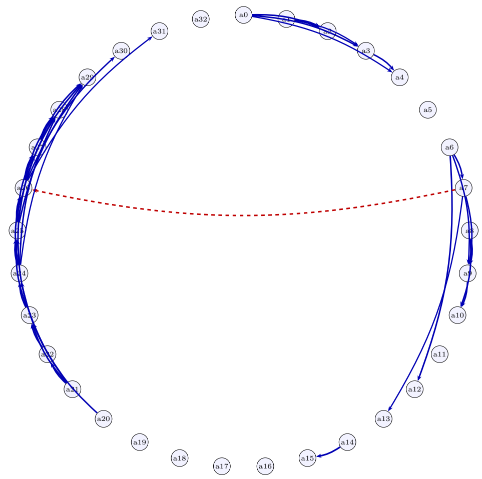
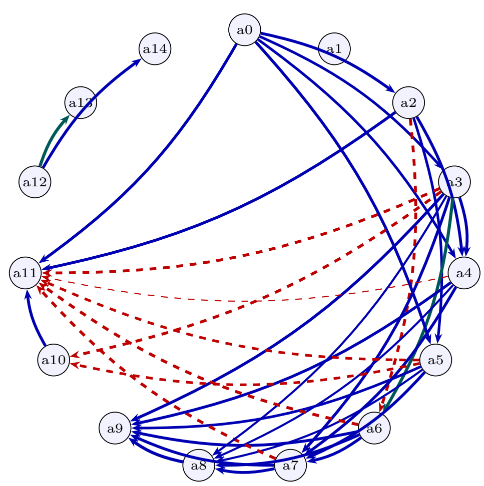


Figure 8: Relation graph for `e2-con` (Biography (with planted contradictions)): win.



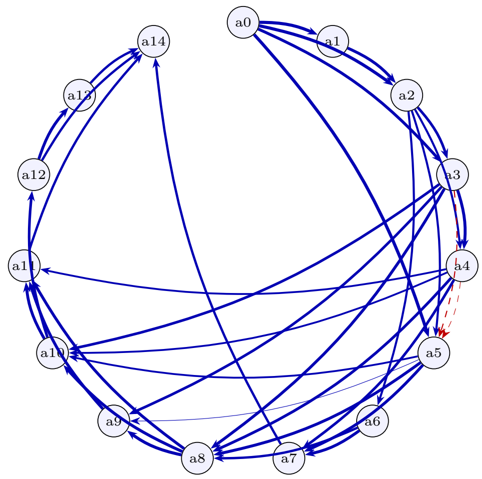
win: 40 relations

Figure 9: Relation graph for `e3-nar` (Narrative passage (Elinor)): win.



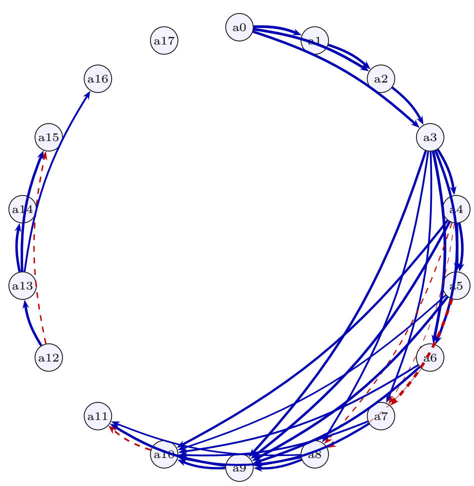
win: 36 relations

Figure 10: Relation graph for `e4-sum` (Synthesized summary S (reliable + unreliable sources)): win.



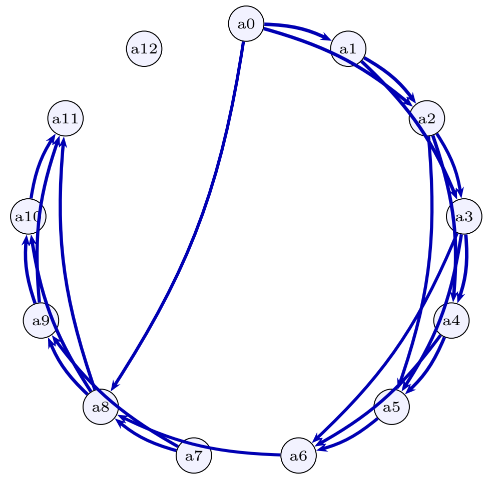
win: 36 relations

Figure 11: Relation graph for `e5-K` (R v Renda summary K (faithful natural ordering)): win.



win: 36 relations

Figure 12: Relation graph for `e5-S` (R v Renda summary S (self-serving-first ordering)): win.



win: 23 relations

Figure 13: Relation graph for `ex6-incident` (Software incident post-mortem (coherent)): win.

models and conditional-strength uncertainty-quantification methods. Three findings are robust. First, the choice of conditional-strength UQ method (surrogate-token from logprobs, sampled affirm fraction, or the verbalized baseline) materially changes the mined edge weights and therefore the LCS, confirming that the strength estimator is a first-class design decision rather than a detail. Second, the four readouts agree on the ordering of coherent versus contradiction-bearing responses but differ in dynamic range, so the headline mean-marginal score is best reported alongside the log-partition diagnostic. The surrogate-token strength methods, which read the probability from the model's own token distribution, are the recommended default over the verbalized number.

Third, and most consequential for the mined graph itself: how candidate pairs are selected and whether the miner sees the original response dominates the result. Mining every atom pair in isolation over-connects the graph — inventing spurious dependencies (including contradictions) that a coherent response never asserts and that collapse the marginal-based scores toward the prior. Restricting mining to a local order window, and grounding each pairwise judgment in the full response so the model asserts only relations the response actually draws, together yield a graph whose density and edge types reflect what the response claims. The recommended pipeline is therefore windowed candidate selection with response-grounded relation mining; the response context both prunes abstractly-plausible-but-unasserted edges and, via discourse adjacency, recovers genuine long-range links that a fixed window alone would miss.

9 Future work

Several directions follow naturally.

- **Calibration on labeled relations.** Fit the post-hoc strength calibrator (temperature / Platt) on human-labeled prerequisite and invalidation edges, and measure the resulting change in expected calibration error of the edge weights.

- **Larger model and method sweep.** Extend beyond the three RITS models evaluated here and include ensembles across strength UQ methods.

- **Human-rated coherence correlation.** Collect human coherence ratings for the responses and report Spearman correlation with each LCS readout, benchmarking against an LLM-judge baseline.

- **Validating grounded recall on long responses.** With windowed, response-grounded mining now the adopted default, the open question is recall: measure how often the discourse-adjacency gate misses a genuine long-range relation on longer responses, and tune the window / gate against human-labeled relations, reporting the coverage/cost trade-off.
- **Joint factuality and coherence.** Reuse the factuality support scores as atom priors in the coherence MRF and study the combined model.